

Realize norm-conforming functional safety

Norm situation: functional safety

Machine industry

Process industry

Electric
Hydraulic
Pneumatic
Mechanical

IEC 62061

IEC 61511

EN ISO 13849

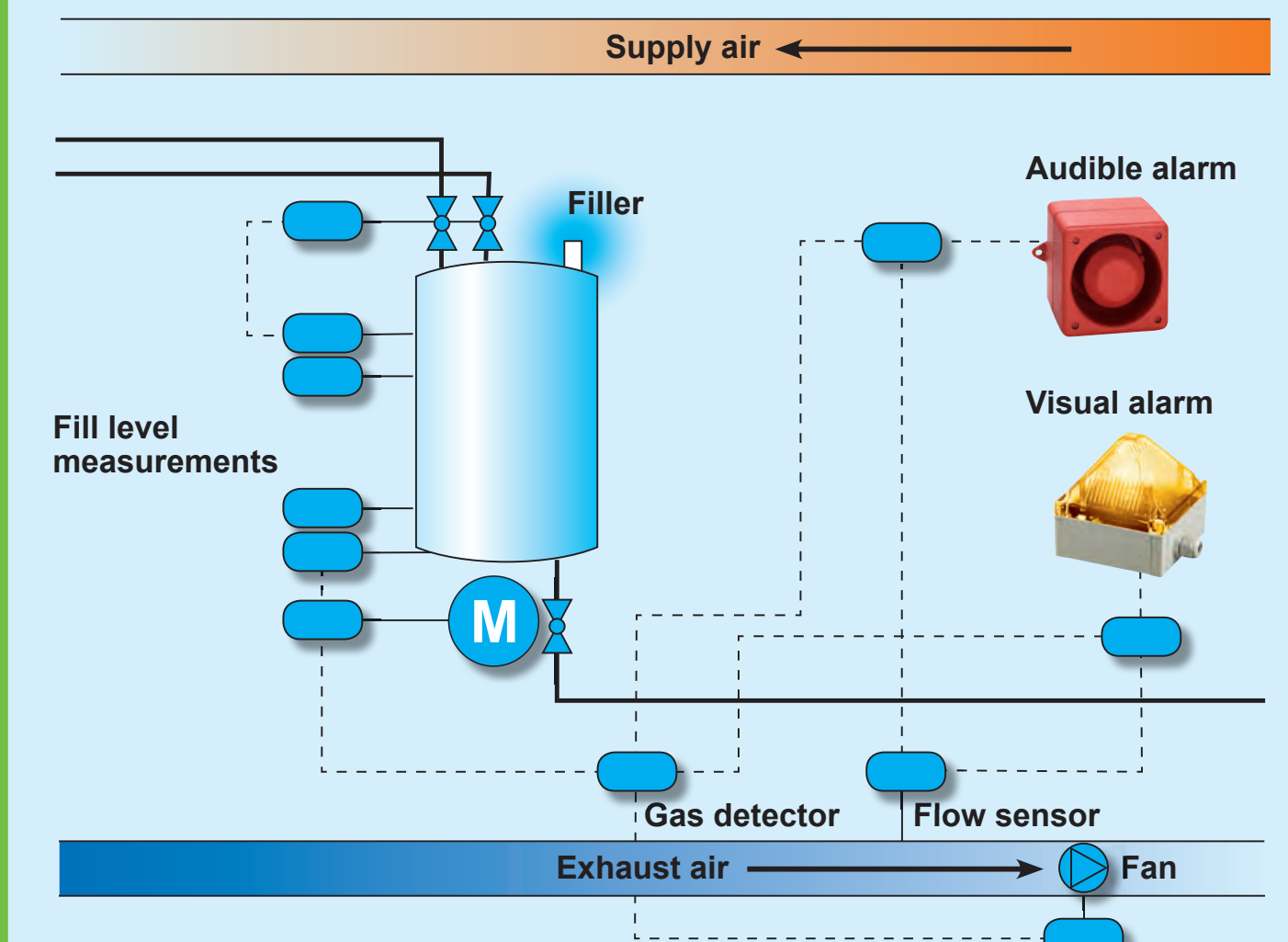
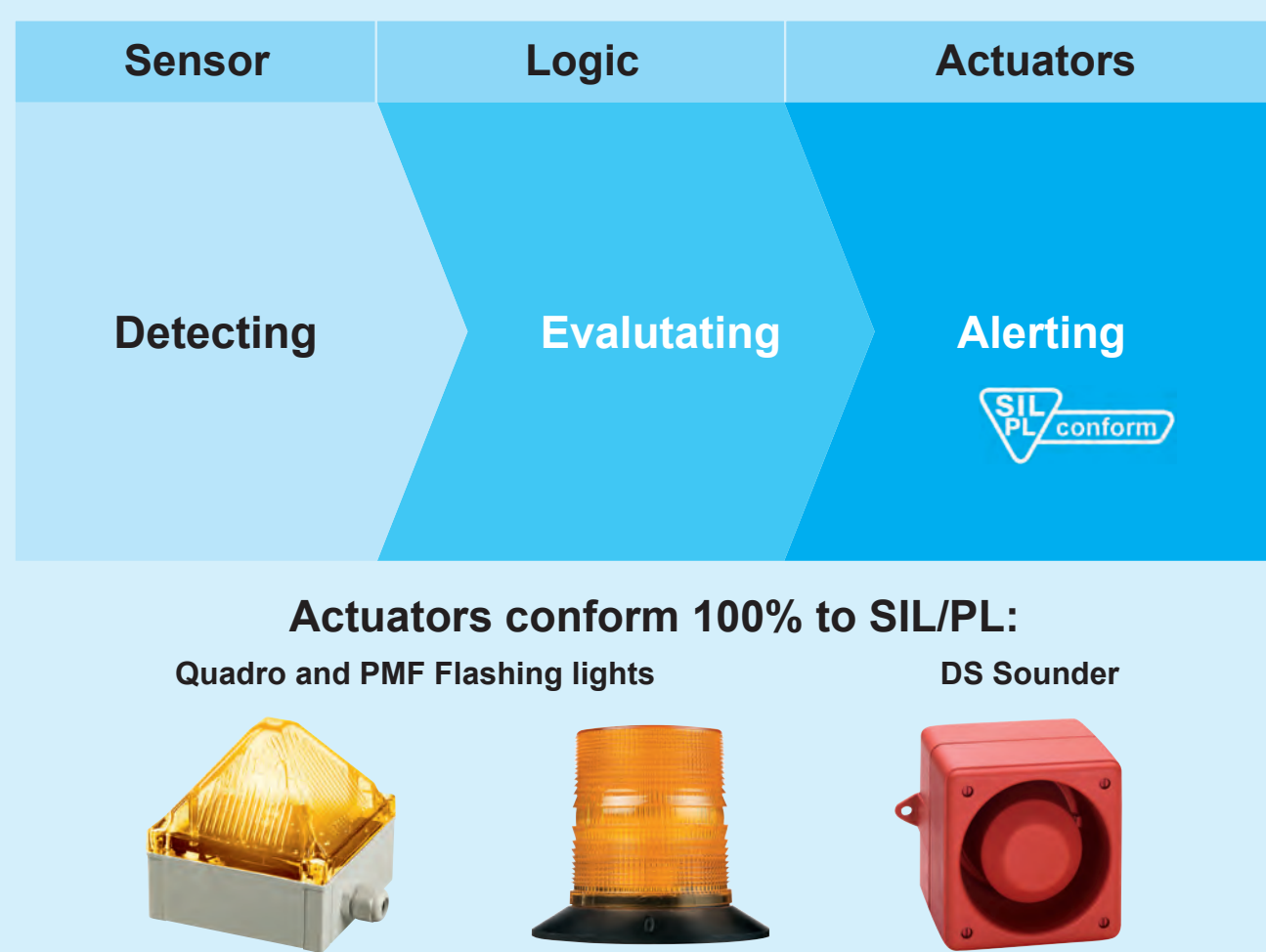
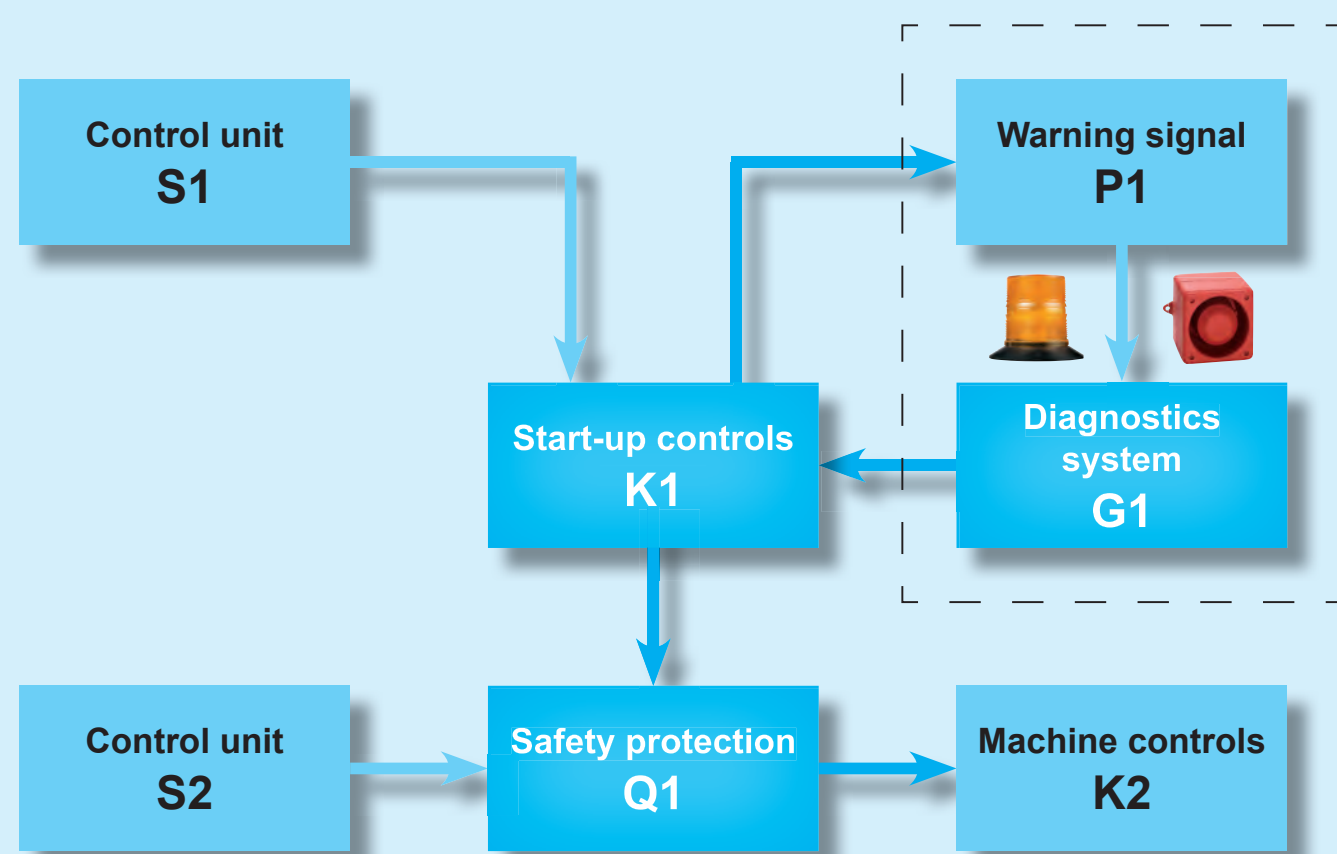
IEC 61508

Electric
Electronic
programm. electronics
(E/E/PE)

Machine safety
e.g. start-up warning

Safety Loop

Process safety
e.g. gas leak alarm

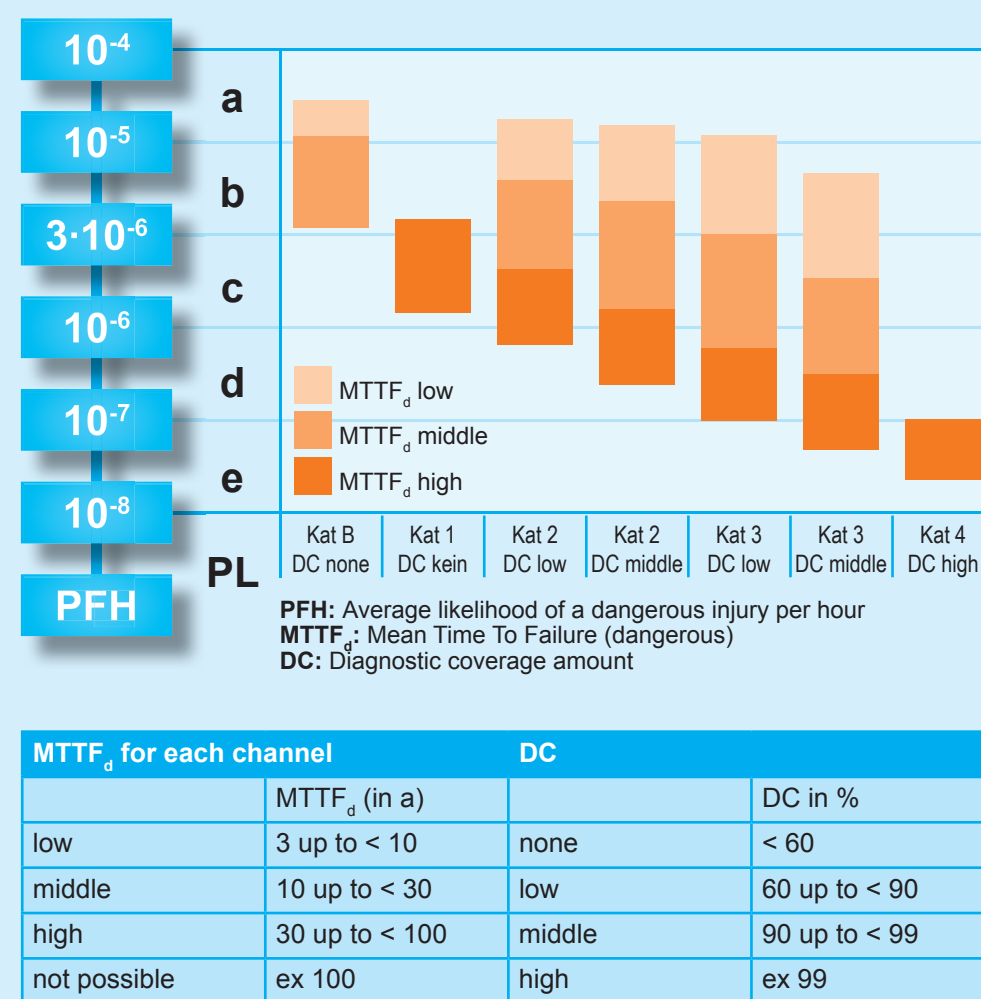
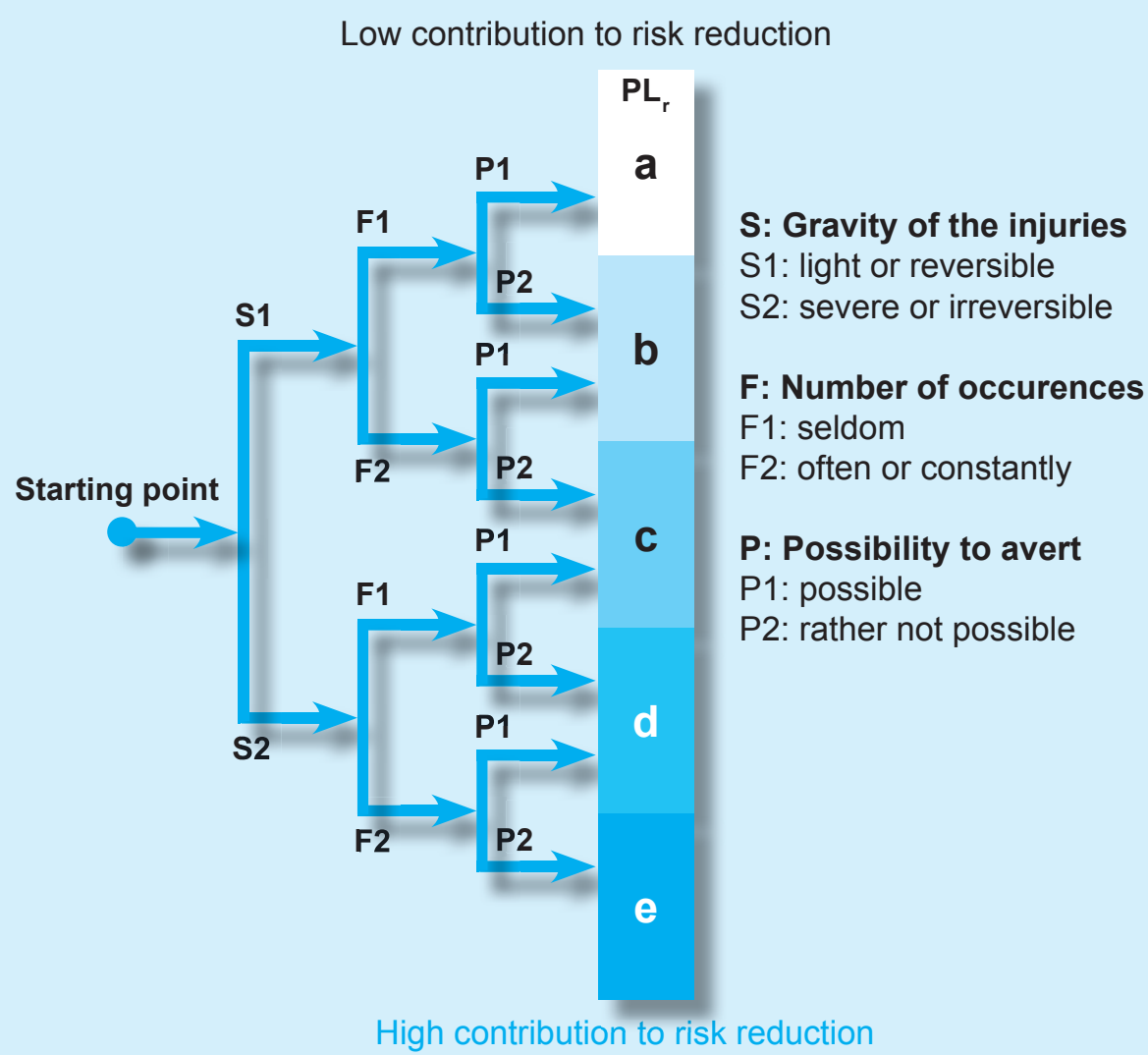


Risk factor according
to DIN EN ISO 13849

Performance Level (PL)
according to DIN EN ISO 13849

Characteristics

Safety integrity, norm comparison,
PFH, PFD, requirements



Portion of safe failures (SFF),
Diagnostic coverage amount
(DC)

$$SFF = \frac{\lambda_{s, \text{without diagnostic}}}{\lambda_{\text{ges}}} \quad SFF = \frac{\lambda_{s, \text{with diagnostic}}}{\lambda_{\text{ges}}}$$

$$DC = \frac{\lambda_{dd}}{\lambda_{dd} + \lambda_{du}} \quad SFF = \frac{\lambda_{s, \text{with diagnostic}}}{\lambda_{s, \text{with diagnostic}} + \lambda_{dd} + \lambda_{du}}$$

$$\lambda_{\text{ges}} = \lambda_{s, \text{with diagnostic}} + \lambda_{dd} + \lambda_{du}$$

s: safe
d: dangerous
dd: dangerous detected
du: dangerous undetected

Safety integrity (type B) according to IEC 61508			SIL/PL (ISO 13849)	
SFF	HFT		SIL	PL
< 60%	–			
60% up to < 90%	SIL 1	SIL 2	1	b, c
90% up to < 99%	SIL 2	SIL 3	2	d
99% up to < 99.9%	SIL 3	SIL 4	3	e

Requirements according to IEC 61508, Type B (in part, unknown failure characteristics)
Comparison SIL/PL (IEC 61508/ DIN EN ISO 13849)

SIL	PFH(d)	PFD(d)	Characteristics (IEC 61508)
1	< 10 ⁻⁵	< 10 ⁻¹	SIL Safety Integrity Level
2	< 10 ⁻⁶	< 10 ⁻²	SFF Portion of safe failures
3	< 10 ⁻⁷	< 10 ⁻³	PF Failure likelihood
4	< 10 ⁻⁸	< 10 ⁻⁴	PFH PF per hour

Lexicon A-P

Hierarchical arrangement of the EN Norms

Lexicon R-Z

β (beta-Factor or rather, Common Cause Factor)
Measure for the CCF; portion of failures, which have a common cause.

CCF (Common Cause Failure)
Failure due to common cause.

DC (Diagnostic Coverage)
Measure for the effectiveness of the diagnostic, which can be defined as the relationship of the failure rate of the recorded dangerous failures and the failure rate of the total dangerous failures.

DC_{avg}
Average diagnostic coverage.

HFT (Hardware Failure Tolerance)
Ability of a SRECS of a system or system element to complete a required function during the presence of a failure or breakdown.

KAT (Category)
Setting of the safety-related components of the controls with relation to their resistance against failures and the respective behavior following, which is attained according to the structure of the component alignment, the failure recognition and/or their dependability.

λ
Average probability of a failure.

λ_d
Rate of dangerous failures.

λ_s
Rate of safe failures.

MTTF_a (Mean Time To Dangerous Failure)
Average time / mean time to a dangerous failure.

Muting
By-pass function: a compliant time-limited override of the safety function with additional sensors.

PFH/PFH_a (Probability of [Dangerous] Failure per Hour)
(dangerous) failures per hour during continuous use.

PFD (Probability of Failure per hour on Demand)
Failure probability when safety function is triggered / activated.

PL (Performance Level)
Discrete Level, which specifies the ability of safety-related control components to execute safety function under predictable conditions.

PL_n (Performance Level, necessary)
To attain a necessary risk minimization for safety functions.

Excerpt of the relevant norms for the use
of visual and acoustic signaling units

Basic safety norm
e.g. EN ISO 12100

Safety group norm
e.g. EN ISO 13849-1
EN 62061

Engineering standard

Type A Norms
General design guidelines and basic definitions for machines

Type B Norms
For example:
EN 61310
EN 981
EN 842
EN ISO 7731
Indication, marking and actuation
System of acoustic and visual alarm signals and information signals
Visual danger signals
Alarms for public areas and workplaces

Type C Norms
For example:
EN 15093/15094
EN 415
EN ISO 11111
EN 1010
Rolling machines
Packaging machines
Textile machines
Printing machine

EN 692/693
EN 1034
EN 1248/1247
Presses
Paper machine
Casting machine

Risk
Combination of the probability of loss / damage occurrence and the extent of the damage.

SFF (Safe Failure Fraction)
Portion of safe failures, portion of the total failure rate of a sub system, which does not cause a dangerous failure.

Safety function
Machine function, which if fails, automatically increases the risk (the risks).

SIL (Safety Integrity Level)
Discrete step / stage (one of four possible) to specify the safety integrity of the safety function, which is assigned to the E/E/PE-System. The SIL 3 (SIL 4 in the process industry) is the highest step / stage and SIL 1 is the lowest.

SIL_{lim} (SIL-standard limitation)
Maximum SIL, which can be utilized for a SRECS-sub-system with regards to the structural constraints and system safety integrity.

SRCF (Safety Related Control Function)
A SRECS executed control function with a defined integrity level, which is designated to maintain the safe condition of the machine or to prevent the immediate increase of risk.

SRECS (Safety Related Electronic Control System)
Electronic control system of a machine, whose failure immediately increases risk.

SRP/CS (Safety Related Part of Control System)
Portion of a control, which reacts to safety-related incoming signals and generates safety-related outgoing signals.

T_r (Repeat test)
Recurring test in order to detect failures in a safety-related system so that, if necessary, the system can be brought back into a "like new" condition or as close as possible according to the practical factors. Technically speaking, a recurring test is not possible for most units.

T_s (Service life)
Time-span covering the use of the SRP/CS.